

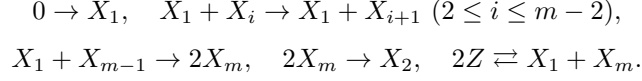
Exact Diffusion Design for Maximally Collective Stable Turing Patterns

in Binary-Complex Mass-Action Networks

Topology-wide all-spectrum localization and exponent-optimal heterogeneity trade-offs

Two-page theorem summary — external audit draft

Family/realization space. For every $m \geq 3$, the binary-complex mass-action topology $\hat{N}_m$ has species $X_1, \dots, X_m, Z$ and reactions



The chain range is empty for $m = 3$. With $n = m+1$, there are n species and $n+1$ reactions. The semipositive conservation vector is $c = (0, 4, \dots, 4, 2, 1)^T$, $\text{rank } \Gamma_m = m$, and

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

Every positive steady flux is $(a\mathbf{1}_m, b, b)$, and every positive-equilibrium Jacobian is exactly $J_m(a, b, H) = A_m(a, b)H$ with $a, b > 0$ and $H \succ 0$ diagonal. Conversely every such triple is physically realized.

Topology-wide localization. For every positive realization and every principal species set I with $|I| < m$, $J_{I,I}$ is Hurwitz. Every nontrivial strongly connected diagonal block is one of two stable long cycles, a principal block of the boundary triad $\{X_1, X_m, Z\}$, or a negative singleton. The triad satisfies the cubic Routh–Hurwitz inequalities coefficientwise. In contrast,

$$(-1)^m \det J_{\{X_1, \dots, X_m\}, \{X_1, \dots, X_m\}} = -2a^{m-1}b \prod_{i=1}^m h_i < 0.$$

Hence, throughout every positive realization,

$$\min\{|I| : \alpha(J_{I,I}) > 0\} = m = n - 1.$$

Every smaller subsystem is stable against both real and complex instability.

Principal-minor diffusion-ray theorem. Let $a_I = (-1)^{|I|} \det J_{I,I}$. Assume $\det J = 0$, $a_I > 0$ for $|I| \leq n-2$, and $\sum_{|I|=n-1} a_I > 0$ (in particular these coefficient hypotheses follow from a simple zero and otherwise stable spectrum together with stable lower principal blocks). Then for $D \succ 0$

$$\det(sD - J) = s(\beta_1 + \beta_2 s + \dots + \beta_n s^{n-1}), \quad \beta_k > 0 \quad (k \geq 2).$$

A nonzero stationary threshold exists iff $\beta_1 < 0$; it is unique and simple. Moreover $J - sD$ has a positive real eigenvalue exactly for $0 < s < s_*$. This does not rule out nonreal wave instability outside that band. The network application is a one-bad-minor corollary, not a hypothesis of the general theorem. Here scalar simplicity is first proved for the dispersion root; ordinary algebraic simplicity of zero for $J - s_*D$ follows separately from $\chi'_{s_*}(0) > 0$.

Exact network diffusion law and sharp linear cost. The complete order- $(n-1)$ omission table has one negative term (omit Z), positive weight 16 when an interior X_j is omitted, and zero when X_1 or X_m is omitted. Thus

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j}$$

is necessary and sufficient for a stationary threshold on the ray sD . For every H in the explicitly defined homogeneously stable domain, the sharp nonattained contrast infimum is

$$T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1};$$

at $H = I$ it is $8(m-2)$. Every crossing obeys $\chi_D \chi_H > 8(m-2)$.

Exact unit-equilibrium stable design. At $a = b = 1$, $H = I$, write $K_i = 91m - 181 - i$ and take

$$d_1 = \frac{23}{63}, \quad d_i = K_i^{-1} \quad (2 \leq i \leq m-1), \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}.$$

The exact contrast is $23(91m - 183)/63$. A 35-term homogeneous half-plane certificate and a 77-term spatial certificate prove a primary simple first-mode zero with all other modes stable. Exact critical vectors give transversality $\eta_m > 0$. The homogeneous base case $m = 3$ is checked from $q_3 = (\lambda + 7)(\lambda^2 + 5\lambda + 2)$; the 35-term root exclusion applies for $m \geq 4$.

On the fixed integrated-mass class, the quadratic field satisfies

$$A_m w_0 = -\frac{1}{4}B(r, r), \quad c^T w_0 = 0, \quad (A_m - 4D_m)w_2 = -\frac{1}{4}B(r, r).$$

The cubic coefficient is

$$c_m = \frac{\ell^T \{B(r, w_0) + \frac{1}{2}B(r, w_2)\}}{\ell^T r}.$$

A four-factor recurrence and a shifted-polynomial certificate prove $c_m < 0$ for every $m \geq 3$. The two supercritical positive branches are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.

Stable diffusion-equilibrium trade-off. Let $\nu = m - 2$,

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \quad L_0 = \kappa_\nu/\sqrt{\nu}, \quad L_1 = 90\nu/(90\nu + 1),$$

and for $L \in [L_0, L_1]$ set

$$h_1 = h_m = h_Z = 1, \quad h_i = \frac{K_i}{LK_{i-1}}, \quad D^{\text{phys}} = H\Delta,$$

where Δ is the effective unit profile above. The physical equilibrium is $H^{-1}\mathbf{1}$. The interval is nonempty in every dimension. A 22-term homogeneous certificate for $\nu \geq 2$, a direct Routh–Hurwitz calculation for $\nu = 1$, an 84-term spatial certificate, and a gauge-corrected cubic comparison prove a primary supercritical branch that is locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space for every certified L . The physical contrasts are

$$\chi_D(L) = \frac{23}{63}91\nu L, \quad \chi_H(L) = \frac{91\nu - 1}{91\nu L}, \quad \chi_D\chi_H = \frac{23(91\nu - 1)}{63}.$$

At $L = L_0$,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \quad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu} \frac{91\nu - 1}{91\nu},$$

so both are $\Theta(\sqrt{m})$. The lower endpoint is the boundary of the present sufficient homogeneous modulus estimate for $\nu \geq 2$; it is not a proved intrinsic dynamical boundary. Since every crossing satisfies $\max(\chi_D, \chi_H) > \sqrt{8(m-2)}$, exponent $1/2$ is globally optimal. No constant-optimal or complete Pareto-frontier claim is made.

PDE conclusion. On the real fixed-integrated-mass H^1 space, positive Neumann diffusion is sectorial and the quadratic mass-action Nemytskii map is smooth into L^2 . The conservation restriction removes the homogeneous zero and the Neumann interval has no continuous translation mode. Strict branch-spectrum negativity therefore yields local exponential convergence for sufficiently close nonnegative initial data in the same mass class.

Most failure-prone points. External checking should focus on: (i) exhaustion of principal SCCs, including $b = 2a$; (ii) strict equality cases in the 22/35/77/84-term certificates and the direct $\nu = 1$ cubic; (iii) the principal-minor derivative monotonicity; (iv) the equilibrium-scaled fixed-mass gauge; and (v) the cubic comparison $N_m(L) > 1/200$.

Scope. The family is synthetic, rank deficient with one semipositive conservation law, and all-mobile. Results are local for each fixed dimension. No arbitrary-data global boundedness, global attraction, constant-optimal stable frontier, biochemical realization, cross-diffusion, or immobile-species theorem is claimed.